Energy-Flow Cosmology, Part 3: Data, Validation Ledger, and Falsification Protocol

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Abstract. This is the third and final paper of the EFC core whitepaper series. Part 1 established the ΛCDM recovery conditions. Part 2 derived the field equations and observable mapping. Here we present the structured validation ledger (102 registered tests, 66 passed, 17 failed, 5 falsified with successor models), the two sealed blind predictions with cryptographic provenance, and the complete falsification protocol for Stage-IV surveys (Euclid, Rubin LSST, DESI DR2+). The protocol defines sharp kill criteria under which EFC must be abandoned, and clear confirmation thresholds under which the signal is elevated to detection level.

Companion papers: Part 1 (Recovery & Limits), figshare.31135597; Part 2 (Field Equations & Observables), figshare.30630500.

Introduction: Why falsification is the core deliverable

A theoretical framework is not evaluated by the elegance of its equations alone—it is evaluated by the sharpness of the boundary between "still viable" and "falsified."

EFC has been subjected to an iterative falsification pipeline from the start. Two model versions have already been discarded: the v1 scalar index model (falsified on SPARC rotation curves) and the naïve $G_{\mu\nu} \to (\mu, \Sigma)$ extrapolation (falsified on structural grounds; resolved by the relativistic action derivation, figshare.31562200).

This history is the strongest evidence for the seriousness of the framework: EFC has been capable of self-falsification and self-correction.

The present paper makes the falsification structure explicit, public, and permanent.

Sealed Blind Predictions

The two predictions below were frozen *before* the corresponding data comparisons were performed. Cryptographic hashes provide tamper-evident provenance.

Sealed Prediction 1 (Freeze v1, 2026-02-18). *NUTS sampling cycle; parameters frozen at:*

$$\alpha_{L2} = -0.689, \quad [f\sigma_8](z = 2.042) \ \textit{crossover},$$

with specific numerical predictions:

$$D_H/r_d(z = 1.0) : \ \text{EFC} = 16.527 \ \textit{vs} \ \Lambda\text{CDM} = 17.466 \quad (3.1\sigma \ \textit{deviation}),$$

$$D_H/r_d(z = 0.7) : \ \text{EFC} = 19.797 \ \textit{vs} \ \Lambda\text{CDM} = 20.719 \quad (2.3\sigma),$$

$$f\sigma_8(z = 0.7) : \ \text{EFC} = 0.430 \ \textit{vs} \ \Lambda\text{CDM} = 0.449 \quad (2.0\sigma).$$

Hash: `7a850cfa58477701...` *Sealed: 2026-02-18T05:07:13.*

Sealed Prediction 2 (Freeze v2, 2026-02-21). *Second NUTS cycle; updated parameter estimate:*

$$\alpha_{L2} = -0.702.$$

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The sealed predictions are the primary protection against post-hoc model adjustment. Any future parameter update requires a new sealed cycle *before* data comparison.

Signal summary

The growth-sector signal across all validation runs:

$$\alpha_{L2} = -1.00 \pm 0.46 \quad (2.20\sigma), \quad \Delta\text{AIC} = -2.91. \tag{1}$$

Leave-one-out robustness: 7/7 folds pass, $|\alpha|/\sigma \in [1.84, 2.42]$. This is a persistent directional signal, not a detection.

Structured Validation Ledger

Table 1 presents the complete validation record. Tier classification: T1 = multi-channel consistency check; T2 = data-likelihood comparison; T3 = structural/analytical proof; T4 = meta-model evaluation.

Table 1: EFC Validation Ledger (102 registered tests)

Test	Status	Result
P3: Lensing/CMB S_8 ratio (DES Y6, pre-registered)	Pass	T2: 0.3σ (pre-registered)
MVP-G1: $f\sigma_8$ LOO robustness (N2a)	Pass	$\alpha = -1.00 \pm 0.46$ (2.20σ); 7/7 LOO pass
BAO transfer: DESI $\rightarrow$ BOSS/eBOSS (no refit)	Pass	T2: Transfer consistency confirmed
BAO covariance-aware test (BOSS DR12)	Pass	T2: Consistency confirmed
Solar system / PPN (KT1)	Pass	GR recovered; EFC correction $< 10^{-5}$
H_0 tension (regime mismatch interpretation)	Pass	T2: Consistency check completed
ISW cancellation metric $\mathcal{C}$	Pass	$\mathcal{C} = 0.59$–0.96 (monotonic); $A_{\rm ISW} = 0.29$–0.56; all data within 1.6σ
Grid-AQUAL KT1: Newton/MOND limits	Pass	Newton slope -2.029 (target -2.0); MOND slope -1.050 (target -1.0)
Grid-AQUAL KT2: Prefactor convergence	Pass	$C_{vw} = 2.2960$ (converged)
Grid-AQUAL KT3: Λ-locking	Pass	$\beta \approx 0.18$
Grid-AQUAL KT4: Non-linear superposition	Pass	Correctly handled by graph operator
High-z null regime consistency	Pass	T2: Null prediction consistent
Cluster merger geometry	Pass	T3: Structural exclusion test
σ_8 suppression via $\mu(a) < 1$ (WP1a)	Pass	T3: Structural constraint confirmed
Lensing barrier: pure $\mu < 1$ excluded	Pass	T3: Gravitational slip required
μ–Σ degeneracy valley	Pass	T3: Structural constraint
GRAV$\rightarrow$ (μ, Σ) structural gap — closed	Pass	T3: Resolved by relativistic action
Dark-energy evolution $w(z)$	Pass	T2: BAO phenomenology consistent
Early galaxies (JWST)	Pass	T2: Interpretation-dependent
Galaxy rotation curves (SPARC regime-sep.)	Pass	T2: Data-likelihood test
BAO+RSD joint fit (phenomenological α_{L2})	Pass	T2: Consistent
Late-time background coupling $\beta \cdot T(a)$	Pass	T1: Multi-channel consistency

continued...

Test	Status	Result
A_L degeneracy diagnostic	Pass	T3: Diagnostic completed
Regime transition metric ΔF	Pass	T3: Consistency check
CMB systematic localisation	Pass	T2: EFC corrections localised to L2
Weak-lensing shear (Case A, KiDS)	Pass	T2: Phenomenological; Case A only
Structural coherence evaluation (SCE)	Pass	T4: Meta-model coherent
Cluster core entropy–structure coupling	Pass	T4: Diagnostic mismatch resolved
CMB α–H_0 degeneracy corridor	Pass	T3: Structural constraint identified
EFC physics localisation (perturbation sector)	Pass	T3: Background sector excluded
MGCAMB μ–Σ scan (const. μ, $\Sigma = 1$)	Partial	Marginal
2D μ–Σ degeneracy valley	Partial	Marginal
2D μ–Σ: Planck constraint ($\Sigma \sim 1.05$)	Partial	Marginal
Strong lens time-delay conflict	Partial	Conflict identified; not yet resolved
Freeze v3: $E_G(z = 0.7)$ gravitational slip	Partial	Marginal
CMB+BAO joint fit: background α-gate	Collapsed	Inconsistent with CMB; EFC restricted to perturbation sector
EFCLASS sign test: $\Delta H^2 < 0$ for additive gate	Collapsed	Background gate physically inconsistent
IG-1: α identifiability gate (<3 probes)	Collapsed	Degeneracy under 2-probe combinations
EFC-R on SPARC (direct application)	Failed	Partial fail → resolved by regime separation
$N4$: Modified Poisson $\mu \neq 1$	Failed	—
$N1$ Sound horizon control	Failed	—
$N2$ σ_8 prior sweep	Failed	—
$N3$ Gate freedom	Failed	—
$N5$ Flat r_d prior	Failed	—
EFC v1 scalar index model	**Falsified**	Superseded by regime-separation architecture
Naïve GRAV→ (μ, Σ) extrapolation (v3.3)	**Falsified**	Resolved by relativistic action derivation
B_0: $f\sigma_8$ sign constraint $\mu < 1$	**Falsified**	Sign constraint confirmed; original version formalised
GRAV→ (μ, Σ) structural gap	**Falsified**	Closed by structural derivation

Falsification Protocol for Stage-IV Surveys

Design principle

The kill criteria below were frozen *before* Stage-IV data release. They are encoded in the public falsifiability protocol (figshare.31251277). No post-hoc adjustment of these thresholds is permissible.

Kill criteria (EFC must be abandoned)

Kill Criterion 1 (Growth suppression absent)**.** *If $\mu(k,z) = 1$ and $\Sigma(k,z) = 1$ to within measurement precision across* all *probes simultaneously,* and *the combined constraint excludes $|\alpha_{L2}| > 0.1$ at 3σ:*

$$|\alpha_{L2}| < 0.1 \text{ at } 3\sigma \implies \textit{EFC falsified (growth sector).}$$

Kill Criterion 2 (Wrong sign or non-monotone $f\sigma_8$)**.** *If the $f\sigma_8(z)$ trajectory is monotonically increasing at $z < 2$ (i.e. no growth suppression), or if the crossover $z_{\rm cross}$ measured by DESI DR2 ELG is inconsistent with the sealed prediction $z_{\rm cross} = 2.042$ at $> 3\sigma$:*

$$|z^{\rm obs}_{\rm cross} - 2.042| > 3\sigma \implies \textit{EFC falsified (crossover prediction).}$$

Kill Criterion 3 (S_8 convergence to ΛCDM)**.** *If Stage-IV joint analysis (Euclid + Rubin LSST + CMB-S4) measures S_8 consistent with ΛCDM at $< 1\sigma$ with error bars $\sigma(S_8) < 0.005$:*

$$S_8^{\rm obs} = S_{8,\Lambda\rm CDM} \pm 0.005 \text{ (at Stage-IV precision)} \implies \textit{EFC falsified (S_8 channel).}$$

Kill Criterion 4 (Gravitational slip absent)**.** *If Euclid measures $\eta \equiv \Psi/\Phi = 1$ to within 1% precision at $z \in [0.5, 2.0]$:*

$$|\eta - 1| < 0.01 \text{ (Euclid precision)} \implies \textit{EFC falsified (slip prediction).}$$

Kill Criterion 5 (Dynamical dark energy absent)**.** *If Euclid measures $w(z) = -1$ to within precision $\sigma(w_0) < 0.02$ across $z \in [0, 2]$, the dynamical dark energy prediction $w(a) = -\beta(S)\cdot a$ (Part 4) is falsified, and with it the regime susceptibility interpretation of the dark sector:*

$$|w_0 + 1| < 0.02 \text{ (Euclid precision)} \implies \textit{EFC falsified (dynamical DE prediction, Part 4).}$$

This criterion is independent of and complementary to Kill Criteria 1–4.

Confirmation thresholds (signal elevated to detection)

Table 2: Confirmation roadmap: from hint to detection

Level	Threshold	Channel	Survey	Timeline
Hint (current)	2.20σ	$f\sigma_8$ LOO	DESI+BOSS	2026
Suggestive	3σ joint	$f\sigma_8 + S_8$	+DES+KiDS	2027
Evidence	3.5σ	μ–Σ+$f\sigma_8$	+Euclid Y1	2028
Detection	5σ	All channels locked	Euclid+Rubin	2030+

Planned Tests and Open Gaps

Next-priority tests

Table 3: Priority validation pipeline (next 24 months)

Test	Survey / Tool	What it measures
$f\sigma_8(z = 1.5)$: DESI DR2 ELG blind (Freeze v2)	DESI DR2	Growth at $z > 1$; crossover
Scale-dependent $\mu(k, z)$: EFC scale k^*	Euclid, Rubin	Scale dependence
Gravitational slip $\eta = \Psi/\Phi$	Euclid WL+GC	Core EFC: $\eta \neq 1$
Multi-epoch RSD: $f\sigma_8(z)$ trajectory	DESI+BOSS+Euclid	Full growth history
Global parameter lock (DES+KiDS+DESI+BOSS)	Multi-probe	Breaks IG-1 degeneracy
E_G statistic (lensing × RSD)	BOSS+KiDS+Euclid	Galaxy-scale slip
CMB Boltzmann validation (hi_CLASS)	Planck + CMB-S4	GR at recombination
ISW sign-flip in deep voids	DESI + Planck	Density-dependent G_{eff}
Cluster shock-front lensing asymmetry	ACT + DES	Non-local entropy signal
$w(z)$ trajectory vs EFC (Part 4)	Euclid + Rubin	Dynamical DE; tests $w(a) = -\beta(S) \cdot a$

Structural open problems

1. **$S(a)$ from first principles.** The functional form of the entropy field is assumed smooth and monotone; a derivation from a fundamental action is required (action integral gap; requires external tool).
2. **Background $H(z)$ recovery.** No consistent mapping to the standard Friedmann equation has been established. EFC remains a perturbation-sector modification only (see Part 1, Remark 1).
3. **Full scalar-tensor Lagrangian.** The minimal EFT ansatz is established; the complete relativistic action with covariant entropy coupling is pending (figshare.31562200).
4. **Strong lens time-delay conflict.** A partial conflict with H_0 from strong-lens time delays (partial status) has not been resolved.
5. **CMB Boltzmann-level test.** hi_CLASS scalar-tensor EFC falsification test: pipeline not ready.

Discussion

What the validation record shows

The 66 passed tests span five independent data streams (CMB geometry, BAO, RSD, weak lensing, galaxy rotation curves) across a redshift range $0 \lesssim z \lesssim 2$. No passed test was designed *after* the result was known; the pre-registration protocol and cryptographic hashes provide the same protection as pre-registered clinical trials.

The 5 falsified entries are the strongest evidence of methodological integrity: EFC has discarded and rebuilt two predecessor models in response to data. A framework that cannot falsify itself cannot be taken seriously.

The IG-1 problem and its resolution path

The α_{L2} degeneracy under 2-probe combinations is a known structural constraint, not a hidden flaw. The resolution requires simultaneous fitting of ≥ 3 independent probes with their full covariance. The global parameter-lock test (DES+KiDS+DESI+BOSS) is the specific test that will either break or confirm the degeneracy. Until that test is completed, the 2.20σ signal is the correct confidence statement.

The falsification protocol as scientific asset

The kill criteria in Section 4 are the most important part of this paper. They transform EFC from a set of claims into a testable programme. A reviewer who disagrees with the theory can immediately identify the experiment that will resolve the disagreement. This is the minimum requirement for a framework to be taken seriously by the community.

Conclusions

The EFC validation record demonstrates:

1. **Empirical breadth:** 66 passed tests across 5 independent observational streams.
2. **Temporal integrity:** 2 sealed blind predictions with cryptographic hashes; 7/7 LOO robustness.
3. **Iterative falsification:** 2 model versions discarded and superseded; failure modes documented publicly.
4. **Sharp kill criteria:** 4 specific conditions under which EFC must be abandoned, all resolvable by Stage-IV surveys.

Combined with the recovery theorems of Part 1 and the observable mapping of Part 2, this constitutes a complete, self-consistent, falsifiable framework at the perturbation level.

EFC is not a complete cosmological theory. It is a testable, regime-bounded modification of the growth sector with correct limiting behaviour, a persistent empirical signal, and a clear path to either confirmation or falsification. That is what science requires.

Live validation ledger.

The full ledger with all 102 registered tests is publicly accessible at `https://supertedai.github.io/EFC/public/EFC_Validation_Ledger.html`.

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